

1. White Noise Process

A white noise process is defined as:

$$\varepsilon_t \sim WN(0, \sigma^2)$$

such that:

$$E[\varepsilon_t] = 0$$

$$\text{Var}(\varepsilon_t) = \sigma^2$$

$$\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0 \text{ for all } k \neq 0$$

This assumption underpins the residual diagnostics of all ARIMA and ETS models used in this analysis.

2. Differencing Operator

The first difference operator is defined as:

$$\nabla X_t = X_t - X_{t-1}$$

Using the backshift operator B:

$$\nabla X_t = (1 - B)X_t$$

Higher-order differencing is defined recursively as:

$$\nabla^d X_t = (1 - B)^d X_t$$

This transformation is used to induce stationarity in non-stationary time series.

3. ARIMA Model

A general ARIMA(p,d,q) model is defined as:

$$\phi(B)(1 - B)^d X_t = \theta(B)\varepsilon_t$$

where:

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

$$\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$$

$$\varepsilon_t \sim WN(0, \sigma^2)$$

3.1 ARIMA(0,1,0) with drift

This model is:

$$\nabla X_t = c + \varepsilon_t$$

or equivalently:

$$X_t = X_{t-1} + c + \varepsilon_t$$

This represents a random walk with drift.

3.2 ARIMA(1,0,0) model

$$X_t = c + \phi X_{t-1} + \varepsilon_t$$

where $|\phi| < 1$ for stationarity.

3.3 ARIMA(0,1,1) model

$$\nabla X_t = c + \varepsilon_t + \theta \varepsilon_{t-1}$$

4. Portmanteau (Ljung–Box) Test

Hypotheses:

$$H_0: \rho_1 = \rho_2 = \dots = \rho_h = 0$$

(no autocorrelation in residuals up to lag h)

$$H_1: \text{at least one } \rho_k \neq 0 \text{ for } k = 1, \dots, h$$

5. Model Adequacy Conditions

A fitted model is considered adequate if residuals satisfy:

$$E[\hat{\varepsilon}_t] \approx 0$$

$\text{Var}(\hat{\epsilon}_t) \approx \text{constant}$

ACF of $\hat{\epsilon}_t$ shows no significant autocorrelation

These conditions were assessed using residual plots, ACF/PACF plots, and the Portmanteau test.

RESEARCH QUESTIONS

1. How have Total Fertility Rate (TFR) and Total Live Births (TLB) changed over time from 1960 to 2025, and what kind of time series behaviour do they show?

From the initial plots, both series show a clear downward trend, so this question focuses on whether that behaviour is due to a long-term trend or something more stochastic like a random walk. To investigate this, I use differencing, ACF/PACF plots, and stationarity checks.

One difficulty here is that real-world demographic data can be influenced by external factors (e.g. policy changes), so it is not always clear whether patterns are purely statistical or driven by outside effects.

2. Can TFR and TLB be modelled well using ARIMA or ETS models, and which type of model actually fits the data better?

This question comes from the fact that both series show non-stationarity and strong dependence over time. I compare ARIMA models (including differenced models) with ETS models to see which one better captures the structure of the data.

A challenge is that different models can sometimes give similar residual results, making it harder to clearly choose one “best” model.

3. How accurate are the forecasts from these models, and how reliable are they for predicting future values of TFR and TLB?

To answer this, I split the data into training and test sets and compare how well the models perform on unseen data. This helps assess whether the models are actually useful for prediction, not just fitting past data.

A limitation here is that future demographic trends can change due to unexpected events, so even a good model may not always produce highly accurate long-term forecasts.

